VARIANT EVIDENCE · MONTE CARLO

Eric's NF1 Odds of Pathogenicity Calculator

README for `nf1_op_credible_interval.html` – a self-contained browser tool that propagates the uncertainty in every input of the de novo missense evidence model into a credible interval for the odds of pathogenicity (OP) and for ACMG/ClinGen points.

Document date: 2026-11-01 · File: nf1_op_credible_interval.html · Companion files: nf1_op_credible_interval.py, nf1_op_credible_interval_gui.py

1. What the tool does

For one *NF1* missense variant that has arisen *de novo*, the evidence can be expressed as an **odds of pathogenicity**:

$$OP = P(B \mid A) \,/\, P(B \mid \neg A)$$

where **B** is the observation (a de novo missense variant in *NF1*), **A** means that the variant is pathogenic and **¬A** that it is not. Each ingredient is itself only known approximately: proportions come from finite cohorts, prevalence and allele burden are published ranges, and the mutation rate is a model estimate. Instead of a single OP, the calculator draws hundreds of thousands of plausible input sets (Monte Carlo), computes the OP for each, and reports:

- the **median OP** and its **highest-density interval (HDI)**;
- the same for **ACMG/ClinGen points** (a logarithmic rescaling of OP, see section 3);
- the **probability that the points reach at least 8, 6, 4 and 2**;
- how many draws had to be discarded, and a text report that can be downloaded or copied.

2. Quick start

1. Open `nf1_op_credible_interval.html` in a modern browser (Edge, Chrome, Firefox or Safari). Nothing has to be installed, the page uses no external libraries and works offline.
2. Check the inputs in the *Model inputs* panel. The defaults are the illustrative *NF1* example and are identical to the defaults of the Python program.
3. Optional: enter a *random seed* to make a run reproducible, and choose a credible-interval level (95 %, 99 %, 99.5 % or custom).
4. Click **Calculate**. With the default 500,000 draws the result appears in about a second.
5. Read the results panel, then use **Download report** (saves `nf1_op_credible_interval_results.txt`) or **Copy report**. **Reset defaults** restores all inputs and clears the results.

File	Purpose
`nf1_op_credible_interval.html`	This browser calculator (model + user interface in one file).
`nf1_op_credible_interval.py`	Command-line version with the same model (`simulate()`, NumPy). Prompts for each input.
`nf1_op_credible_interval_gui.py`	Tkinter desktop interface that calls `simulate()` from the Python file.
`nf1_op_credible_interval_results.txt`	A saved report (the file name the Download button uses).

3. The model, step by step

For each of the *N* Monte Carlo draws the program samples the five uncertain inputs, combines them, and records the resulting log-OP.

#	Quantity	How it is sampled
1	P(de novo | pathogenic)	Beta($n\cdot\hat{p}$, $n\cdot(1-\hat{p})$) from the point estimate $\hat{p}$ and the cohort size n (binomial sampling uncertainty)
2	P(missense | pathogenic)	Beta($n\cdot\hat{p}$, $n\cdot(1-\hat{p})$), likewise
3	NF1 prevalence	Lognormal: median · exp($z\cdot\sigma$), $z \sim N(0, 1)$, σ = ln(upper bound / median) / 1.96
4	Per-allele de novo missense mutation rate μ	$\mu = 10^x$ with $x \sim N$(mean $\log_{10}$ rate, SD)
5	Rare non-pathogenic allele burden	Lognormal, as for prevalence, with its own median and upper bound

The sampled values are then combined as follows (this is exactly what the code evaluates):

$$P(B \mid A) = P(\text{de novo} \mid \text{pathogenic}) \times P(\text{missense} \mid \text{pathogenic})$$

$$\text{pathogenic de novo missense per individual} = \text{prevalence} \times P(B \mid A)$$

$$\text{non-pathogenic de novo missense per individual} = 2\mu - \text{prevalence} \times P(B \mid A)$$

$$P(B \mid \neg A) = [\ \text{non-pathogenic de novo missense per individual} / 2\ ] \ / \ \text{burden}$$

$$OP = P(B \mid A) \ / \ P(B \mid \neg A) = 2 \cdot \text{burden} \cdot P(B \mid A) \ / \ (\ 2\mu - \text{prevalence} \cdot P(B \mid A)\ )$$

$$\text{points} = 8 \cdot \ln(OP) \ / \ \ln(350)$$

Points. The points scale is the logarithm of OP, scaled so that an OP of 350 (the very-strong level of the Bayesian ACMG/AMP framework) equals 8 points. Every doubling of OP adds **0.95 points**.

Points	1	2	4	6	8
Equivalent OP = $350^{\text{points}/8}$	2.1	4.3	18.7	80.9	350
Usual strength label	supporting	moderate	strong	(between strong and very strong)	very strong

Discarded draws. If a draw gives 2μ − prevalence × P(B | A) ≤ 0, it would imply a negative rate of non-pathogenic de novo variants. Such a draw is discarded and counted; the number and percentage are shown under the results (and in the report). A high percentage means that the inputs are inconsistent with each other and the interval should not be trusted without a second look.

Summary statistics. The median is computed on log-OP and transformed back (median OP = exp(median log-OP)). The **HDI** is the shortest interval containing the requested share of the draws (a window of ⌊credibility × N⌋ consecutive sorted draws with the smallest width). The OP interval is the exponential of the log-OP HDI, so it is skewed to the right; the points interval is computed on the points scale.

4. Inputs

Field	Default	Meaning and accepted values
P(de novo | pathogenic)	0.5	Share of pathogenic variants that arise de novo. Strictly between 0 and 1.
Sample size for de novo proportion	1000	Number of variants/individuals the proportion is based on; sets the Beta spread. Positive whole number.
P(missense | pathogenic)	0.19	Share of pathogenic variants that are missense. Strictly between 0 and 1.
Sample size for missense proportion	3000	As above. Positive whole number.
Median prevalence	0.0004 (1/2500)	Best estimate of NF1 prevalence. Positive.
Prevalence – lower bound	0.000333 (1/3000)	Reported for completeness; **does not set the spread** (see section 8). Positive.
Prevalence – upper bound	0.0005 (1/2000)	Sets the lognormal spread. Must exceed the median.
Mean $\log_{10}$ mutation rate	−4.137773	Per-allele de novo missense rate of NF1 (Samocha et al. 2014, as cited in the program). Any number.
SD of $\log_{10}$ mutation rate	0.054	Uncertainty of the rate on the $\log_{10}$ scale. Zero or larger.
Allele burden – median	0.218	Cumulative allele frequency of rare (AF < 0.01) non-pathogenic NF1 variants (e.g. summed from gnomAD). Positive.
Allele burden – lower bound	0.15	Reported only; does not set the spread. Positive.
Allele burden – upper bound	0.32	Sets the lognormal spread. Must exceed the median.
Monte Carlo draws	500,000	Whole number from 1,000 to 2,000,000. More draws give a smoother, more stable interval.
Random seed (optional)	empty	Non-negative whole number. Same seed and inputs give identical results. Empty = different every run.
Credible interval	95 %	95 %, 99 %, 99.5 % or *Custom…* (50 % up to, but not including, 100 %).

Invalid input is reported in a red box under the buttons and nothing is calculated. Typical messages: *All input fields must contain valid numbers*, *P(…) must be greater than 0 and less than 1*, *Both proportion sample sizes must be positive whole numbers*, *Prevalence: median, lower bound, and upper bound must be positive, and upper bound must exceed median*.

5. Reading the results

Median odds of pathogenicity	95% HDI for OP	Median ACMG/ClinGen points	95% HDI for points
386	[227, 657]	8.13	[7.41, 8.86]



The example above is the result panel for the default inputs and seed 12345 (see section 6).

- **Median OP / median points** – the central estimate. Because OP is skewed, the median is preferred over the mean.
- **HDI for OP / points** – with the default 95 %, 95 % of the plausible values lie inside the interval, and no value outside is more probable than a value inside. Note that the label follows the chosen credible level (for example *99% HDI for OP*).
- **Probability of reaching each minimum threshold** – the share of draws with at least 8, 6, 4 or 2 points. This is the most direct answer to "how sure am I that the evidence is at least this strong?".
- **Status line and footnote** – number of valid and discarded draws.

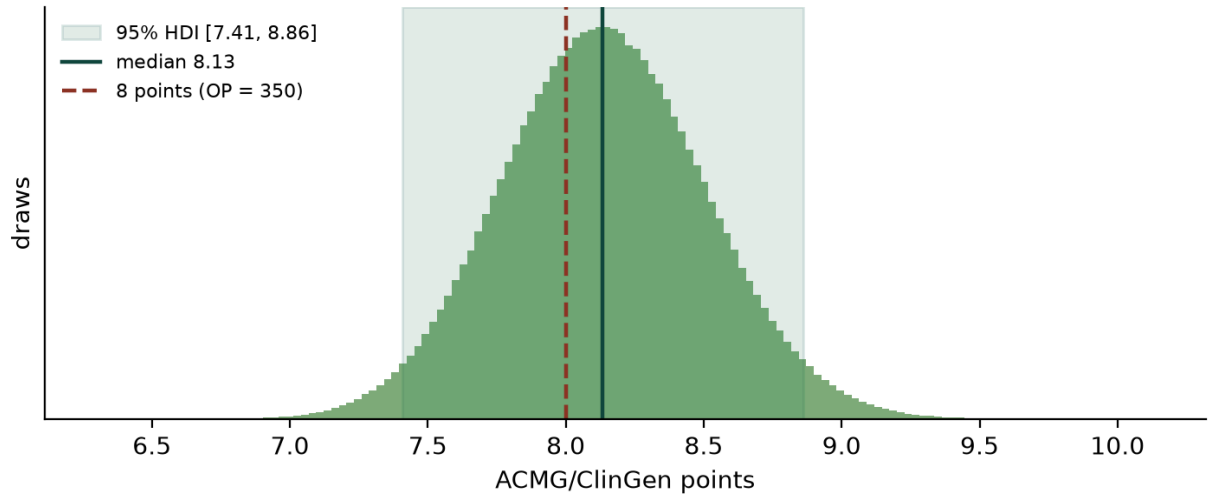
6. Worked example (default inputs)

Default inputs, 500,000 draws, seed 12345, 95 % credible interval. The downloaded report contains the following text:

```
NF1 Odds-of-Pathogenicity (OP) Calculator
Monte Carlo credible interval results

P(de novo | pathogenic): 0.5; n=1000
P(missense | pathogenic): 0.19; n=3000
NF1 prevalence: median=0.0004; plausible range=[0.0003333333333333333, 0.0005]
log10 mutation rate: mean=-4.137773; SD=0.054
Non-pathogenic allele burden: median=0.218; plausible range=[0.15, 0.32]
Monte Carlo draws: 500000; seed=12345
Credible interval: 95%

Median OP: 386
95% HDI for OP: [227, 657]
Median points: 8.13
95% HDI for points: [7.41, 8.86]
P(points >= 8): 64.2%
P(points >= 6): 100.0%
P(points >= 4): 100.0%
P(points >= 2): 100.0%
Accepted draws: 500000
Discarded draws: 0 (0.00%)
```



Distribution of the points for the default inputs (2,000,000 draws, Python reference run; the browser shows the same shape). The shaded band is the 95 % HDI, the solid line the median, the dashed line the 8-point (OP = 350) level.

Interpretation. The best estimate (median 8.13 points, OP ≈ 386) is just above the 8-point very-strong level, but the 8-point line lies *inside* the 95 % interval [7.41, 8.86]: about 64 % of the plausible input combinations reach 8

points, and all of them reach at least 6 (strong plus moderate). The interval therefore supports "at least 6 points" with practical certainty, and "8 points" only with moderate confidence.

Sanity check by hand. With every input at its point estimate: P(B | A) = 0.5 × 0.19 = 0.095; $2\mu = 2 \times 10^{-4.137773} = 1.456 \times 10^{-4}$; OP = 2 × 0.218 × 0.095 / (1.456 × 10^{-4} − 0.0004 × 0.095) ≈ 385, i.e. 8.13 points, in line with the simulated median of 386.

Effect of the credible level

Level	HDI for OP	HDI for points
95 %	[227, 657]	[7.41, 8.86]
99 %	[195, 790]	[7.20, 9.11]
99.5 %	[183, 844]	[7.12, 9.20]

99 % and 99.5 % rows computed with the Python program (1,000,000 draws); the browser gives the same values within Monte Carlo noise.

7. Which inputs drive the uncertainty?

Varying one input at a time (all others fixed at their point estimate) with the default settings gives:

Input that varies	SD of points	Share of total variance
Rare non-pathogenic allele burden	0.27	52 %
Mutation rate ($\log_{10}$ SD 0.054)	0.23	39 %
P(missense | pathogenic)	0.07	3.5 %
P(de novo | pathogenic)	0.06	2.5 %
Prevalence	0.06	2.3 %
All inputs together	**0.37**	100 %

Practical consequences: (1) the width of the interval is dominated by the **allele burden** and the **mutation rate**, so these two inputs deserve the most careful sourcing; (2) OP is directly proportional to the burden, so halving the burden halves OP and lowers the points by about 0.95; (3) larger cohort sizes for the two proportions hardly change the result at these defaults.

8. Notes, assumptions and limitations

- **Lower bounds are not used.** As stated in the tool, the lognormal spread is set only by the median and the *upper* bound, σ = ln(upper / median) / 1.96, to match the Python implementation. The lower bound is kept as an input and printed in the report. Choose the upper bound so that it represents the 97.5th percentile you want to assume.
- **Independent inputs.** The five inputs are sampled independently; no correlation between them is modelled.
- **Beta parameters.** The proportions use Beta($n\cdot\hat{p}$, $n\cdot(1-\hat{p})$), without an added prior count. This is the same in the browser and Python versions.
- **Discarding changes the distribution slightly.** Draws with a non-positive benign rate are removed, so the retained draws are a truncated sample. With the defaults none are discarded; check the footnote when you change the inputs.
- **Seeds.** The browser and NumPy use different random-number generators, so the same seed gives different individual draws in the two programs. Results agree statistically, not draw by draw (see section 9).

- **Scope.** The calculator carries the stated uncertainty through the formulas above. It does not validate the model itself, the input sources, or the clinical rules that apply a given number of points. Interpret the output together with the guidelines of the responsible expert panel.

9. Verification and performance

With the default inputs the browser version and the Python reference function `simulate()` give the same results within Monte Carlo error:

	Median OP	95 % HDI for OP	Median points	95 % HDI points	P(≥ 8)
Browser (500,000 draws)	386	[227, 657]	8.13	[7.41, 8.86]	64.2 %
Python (2,000,000 draws)	386.2	[227.0, 657.9]	8.134	[7.409, 8.862]	64.2 %

Time for the calculation in a Chromium-based browser on the author's PC: about 0.2 s for 100,000 draws, 0.7 s for 500,000 and 3.2 s for 2,000,000 draws (the button is disabled and the panel dimmed while it runs). Doubling the number of draws roughly doubles the time; a few hundred thousand draws are enough for stable 95 % intervals.

10. Troubleshooting

Symptom	Cause and remedy
“Every simulation was discarded. Check mutation-rate and prevalence inputs.”	2μ is smaller than prevalence × P(B | A) in all draws. Check the mutation rate (it is entered as $\log_{10}$, so −4.14 is 7.3×10^{-5}), the prevalence and the two proportions.
Many draws discarded (footnote)	The inputs are only just consistent. Revisit the mutation rate and prevalence before relying on the interval.
“Clipboard access was blocked by the browser. Use Download report instead.”	The browser refuses clipboard access for local files. Use **Download report** instead.
Results change every time	Normal when no seed is given. Enter a whole number as seed for identical results.
Page feels slow	Lower the number of draws (2,000,000 is the maximum). 500,000 is the default.
Browser result differs slightly from Python	Different random-number generators; differences should be within Monte Carlo error. Increase the number of draws to compare.